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Recurrent Neural Networks algorithms and applications

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***Abstract*—Since recurrent neural networks (RNNs) were firstly proposed, it is widely used, and many extended RNNs algorithms have been developed, which achieve good results in many application fields. To report the latest research results of RNNs and help researchers quickly understand the latest progress of RNNs algorithm. In this research, we briefly introduce the basic principle of RNN, review more than 20 papers about RNNs and summarize the previous work. Then, we make qualitative and quantitative analyses on four representative algorithms (CNN-RNN, ResNet-110, ResNet-164, and ResNet-1001). Among them, the model of CNN-RNN obtains the best performance on the CIFAR-100 dataset with standard data augmentation. Finally, we discuss the future development trend of RNNs.**

Keywords: Recurrent Neural Networks. review, applications

I. INTRODUCTION

Recurrent neural networks (RNNs) are machine learning tools established by Jeffrey Elman [1]. They take sequence data as input, recurses in the evolution direction of sequence, and a chain connects all cyclic units.

In traditional networks, all inputs (or outputs) are independent of each other. For many tasks it is a very bad idea. However, scholars are inspired by the feedback system of Neurobiology. Jhon Hopfield proposed a network [2], the Hopefield neural network. It is a simple neural network, but it is simple. In 1992, Jeffrey Elman proposed the first fully connected RNN [1]. Compared with traditional networks, RNNs allow neurons in the hidden layer to communicate with each other and store the last output in the hidden layer as information. In other words, it can save more weight the parts previous, that is "Memory". After years of development, models of RNN as follows: Gating mechanism of RNNs, Long Short-Term Memory networks (LSTM) [3], Stacked Recurrent Neural Network (SRNN) [4], bidirectional recurrent neural network (BRNN), etc.

According to the above, RNNs has memory ability, so RNNs is often used in these 4 sides: Text generation: like the question, give the context, and then predict what the words next; Machine translation: it can translate a language into another; Speech recognition: determining the corresponding text according to the input audio. Generate image description: like look at the picture, give a picture, and describe the picture's content. There are some algorithms of RNNs with their application, as shown in Figure 1.

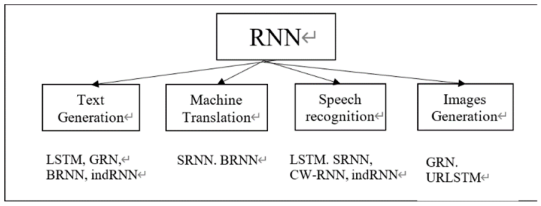

Figure 1. Some algorithms of RNNs with their application.

This paper aims to introduce the extends of RNNs. The remainder of this paper is organized as follows: Section 2 will describe sufficient details of the fundamental of RNN with basic functions. Then, section 3 and section 4 will introduce the representative algorithm and discusses their advantages and disadvantage. Finally, we will combine what we have learned. Then we look forward to the future development of RNNs.

II. RECURRENT NEURAL NETWORK

To understand the principle of RNN, we need to first understand the working process of the biological neural network. The human body is a big machine, and it continuously receives information, process information, and generate output. By decomposing these actions into the biological way, it is uncountable neurons that finish all the tasks. The network of neurons is called a neural network. The structure of a neuron consists of four main parts, and these are dendrite, cell body, axon, and synapses, as shown in Figure 2.

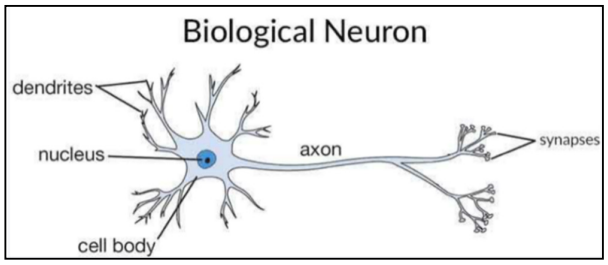

Figure 2. Biological neuron

The RNN structure has an indivisible relationship with it. The function of the dendrite is to receive the signal from the other neurons. The cell body is functions as a terminal to gather all signals input from the dendrite and generate new input. The axon is working as a bridge connecting the cell body and synapses and is a passage for input from the cell body to travel

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down. The synapses are a new beginning of another new neuron, and its function is like a junction. RNN structure basically comes from the biological neural network. However, it is no longer limited by biological arear but became one of the most advanced algorithms in the world. RNN has all the features of a biological neural network. It possesses the input layer, hidden layer, channels, and output layer, as shown in Figure 3.

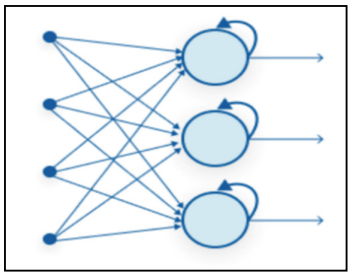

Figure 3. Recurrent neural network

Its unique feature as an algorithm is that it has short-term memory, which enables RNN to has the ability to go back to retrieve previous information. This is the main point to distinguish between RNN and feed-forward neural network. RNN share parameters across the different positions of the sequence. when RNN is decomposed to unfolding model, the formula for computing is

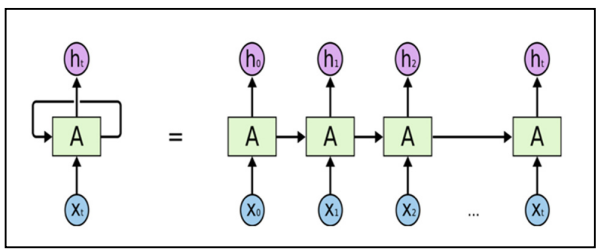

Figure 4. unfolding model of RNN

$$h^{(t)} = f\left(h^{(t-1)}, x^{(t)}; \theta\right), \quad (1)$$

$h^{(t-1)}$ represent the previous hidden state, x^((t)) is current input, and theta represents a parameter of function f. The RNN forward pass can be represented by these equations below:

$$a^{(t)} = b + wh^{(t-1)} + ux^{(t)}, \quad (2)$$

$$h^{(t)} = \tanh\left(a^{(t)}\right), \quad (3)$$

$$o^{(t)} = c + vh^{(t)}, \quad (4)$$

$$\hat{Y} = softmax(o^{(t)}). \quad (5)$$

The Equations (2)-(5) are illustrating a recurrent network that maps an input sequence to an output sequence of the same length. The total loss for a given sequence of x values paired with a sequence of y values would then be just the sum of the losses over all the time steps. We assume that the outputs o(t) areis used as the argument to the softmax function to obtain the vector ŷ of probabilities over the output. We also assume that the loss L is the negative log-likelihood of the true target y(t) given the input so far.

III. REVIEW ON RECURRENT NEURAL NETWORK

We present the related work on Recurrent Neural Network (RNN) as follows: Mao et al. present the multimodal recurrent neural network(M-RNN) used to produce novel image captioning. Its effectiveness is proved on four databases (IAPR TC-12, Flickr 8k, Flickr 30k, and MS COCO), and it is also applied M-RNN to retrieve images or sentences [5]. Jain et al. combine high-level spatial-temporal graphs with Recurrent neural networks to develop an arbitrary spatial-temporal. Then, transforming it into structural-RNN (S-RNN). The S-RNN has a good performance on human motion modeling, human-object interaction [6]. Luo et al. propose a dual-path recurrent neural network (DPRNN). It is a method to organize RNN layers in a deep structure to construct very long sequences. The author experimented on speech separation and recognition to evaluate their approach. DPRNN is effective in the aspect of the acoustic condition and other tasks [7]. Zhao et al. present Hierarchical Structure-Adaptive RNN (HAS-RNN). It is used for structure-adaptive video summarization. The author evaluates their approach on four databases (SumMe, TVsum, Colum, and VTM). Its effectiveness of video summarization has been testified by their experiment [8]. Lv et al. came out with a model called LC-RNN, which intends to predict more accurate traffic speed than the existing solution. By integrating the RNN and CNN model, better time-series patterns of dynamic traffic of surrounding areas come out. The LC-RNN is performing well on the TSP problem [9]. Castrejon et al. presented a semi-automatic annotation of object instances called Polygon-RNN. It can outline the contour polygon of an object. The model authors used is RNN and use CNN representation of image crop as input in each step of RNN. It is a great tool for generic annotation [10]. Li et al. proposed the concept of E-RNN, which is used as FPGA implementation of ASP application. Authors use ADMM-based training block-circulant matrice. It gives out a more efficient RNN framework [11]. Kani proposed E-RNN, which is used as FPGA implementation of ASP ap introduces a concept of deep residual recurrent neural network (DR-RNN) as an efficient tool to reduce nonlinear dynamic systems. Authors combine the POD-based MOR technique with DR-RNN. The statistics show DR-RNN is stable in reduce order [12]. Lin et al. design RNN-SM, which is a novel VoIP steganography in VoIP streams. They compare pruned RNN-SM and full RNN-SM in the experiment. RNN-SM has high detection accuracy even the sample is extremely short [13]. Xiang et al. introduced data-associated recurrent neural networks (DA-RNN), a new frame for 3D scene mapping and labeling semantic. Through experimenting on the dataset from the real-world and RCB-D videos, DA-RNNs can take advantage of 3D scene mapping and video semantic labeling [14].

Chen et al. used Hamiltonian systems and the leapfrog algorithm [15] because Hamiltonian systems can get the relationship between Hamiltonians in the shortest path. Through the leapfrog algorithm of ODEs, the algorithm is more effective than other RNNs in dealing with noise. Simultaneously, optimizing the initial state and the Hamiltonian network presents a hardness of general initial state optimization [16]. Symplectic RNNs, which previous methods in predicting the complex and noisy, more outperforms simulating with the exact equations. The principle of Sliced RNN is slicing the input sequence into several minimum subsequences with equal length. Then the

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information can be transformed into multiple layers and repeat this process until setting minimum value [17]. Compared with standard RNNs, Sliced RNNs faster in dealing with big data.

What's more, the Sliced RNN can do parallel operations, that's mean the longer the sequence length, the faster. Now, Sliced RNN can be used in NLP problems and long sequence tasks. Li et al. proposed design optimization for Efficient RNNs(E-RNNs), aiming to implement an E-RNN framework for ASR application in use FPGA implementations [18]. First, they tried many models, such as LSTM, GRU, and turned off the weight of the model. They found that if the accuracy requirement can still be satisfied, it is desirable to shift from LSTM to GRU because of less computation. Then, they use FPGA platforms for the proposed E-RNN. The results were different FFT of RNN more efficient than ESE and C-LSTM. Jain et, al proposed Structure RNNs(S-RNNs). The approach aims to combine high-level spatio-temporal graphs with sequence [19]. First, the spatio-temporal graph is a good expression of the spatio-temporal relationship, recognizing different elements in the picture with a high recognition rate. First, the spatio-temporal graph is decomposed into factor components, and then the edge feature is used to get the factor graph. Then the SRNN architecture is derived from the factor graph. Finally, a directed graph is generated by RNN connection, which can well divide the boundaries of the objects in the spatio-temporal graph. The model success of RNNs. S-RNN has a good performance in human motion models and human-object interaction. Xiang and Fox et al. proposed Data Associated RNN(DA-RNN). It has a good performance in connecting the analysis of individual RGB-D frames and their integration. Steps main as follows: Firstly, Separating Depth image from RGB image. Then, Label objects with different colors, it can get a rough image as data for the next step. Next, DA-RNN uses the LSTM and GRU to further outline the image to get a more refined label map. Finally, Joint 3D mapping and the semantic label will get 3D scene mapping. In this paper, they said DA-RNN could use in robot vision [20]. Zhang et al. proposed effective EleAttG to empower the neurons in recurrent neural networks to have the attentiveness capability. Then each RNN can have its attention gate, but the cost of computation complexity and number of parameters will largely increase, especially when the dimension of the input is large. EleAtt can be used in any RNN, and if add EleAtt unit, RNN can greatly improve its performance while maintaining accuracy [21]. Merity investigated the research status of natural language processing, the models used, and other alternative methods. Unlike the framework computing power, this paper rethinks whether we can use LSTM and single head attention to complete similar indicators of the large-scale framework on the current dataset. The author created SHA-RNN with LSTM and Attention. Through experiments, the author believes that through careful design, careful adjustment, you can also use the model with less memory consumption to achieve better results [22]. Gaier et al. proposed Weight Agnostic Neural Networks. It claims that it can achieve a good prediction effect without training [23]. Step1: They created an initial population of minimal neural network topologies, step 2: each network is evaluated over multiple rollouts. Step 3, networks are ranked according to their performance and complexity. Step 4, the results of the sequencing are used to regenerate into a new neural network population. Then, repeat from steps 2 to 4. According to experiments, models have good performances in continuous control tasks, and the classification task also has a good performance. Sainath et al. proposed a Full connection model of LSTM and CNN. After inputting the data [26], the model first uses CNN [24] layer to reduce the dimension, then uses LSTM [25] to predict the data, and finally obtains the corresponding prediction according to the data. Compared with LSTM, CLSNN provides 4-6% relative reduction.

Then, we present the details of two important extended RNN algorithms as follows:

A. Clockwork RNN (CW-RNN)

We will summary a paper about Clockwork RNN. Clockwork RNN (CW-RNN) was proposed by Jan Koutn´ık [27]. The LSTM is difficult to train successfully, so Koutn´ık proposed CW-RNN. The hidden layer is partitioned into separate modules. Each processing inputs at its own temporal granularity, making computations only at its prescribed clock rate. The advantage of CW-RNN is it reduces the number of RNN parameters, making train easier to succeed. What's more, its performance is better than LSTM and standard neural networks in speech recognition and audio signal generation.

Hochreiter found that standard RNNs have difficulty learning long-term dependencies encoded in the input sequences due to the vanishing gradient in 2001. LSTM maintains constant backward flow in the error signal. Then, based on an associated error back-propagation through time training algorithm (proposed by Williams in 1989), the authors improved the RNN. They found that not all modules are executed at every time step and have a smaller number of weights than SRNs because slower modules are not connected to faster ones. According to this paper, the CW-RNN had been finished, and it performs well in tasks.

The standard RNN output equations:

$$y_H^{(t)} = f_H\left(W_H \times y^{(t-1)} + W_I \times x^{(t)}\right) \tag{1}$$

$$y_O^{(t)} = f_O(W_O \times y_H^{(t)}) \tag{2}$$

Matrices W_I and W_H as follows:

$$W_H = \begin{pmatrix} W_{H1} \\ \vdots \\ W_{Hg} \end{pmatrix} \qquad W_I = \begin{pmatrix} W_{I1} \\ \vdots \\ W_{Ig} \end{pmatrix} \tag{3}$$

The authors correspond to the executed modules are used for evaluation:

$$W_{Hi} = \begin{cases} W_{Hi} & for(t\ MOD\ T_i) = 0 \\ 0 & otherwise \end{cases} \tag{4}$$

Thus, the low-clock-rate modules process, retain and output the long-term information obtained from the input sequences, which means it is not being distracted by the high-speed modules. Then, the authors set two tasks to train this model. One is Sequence Generation, they found that LSTM, a sliding average, will cause a tiny mistake in a train, but CW-RNN approximates the sequence much more accurately, which means CW-RNN better in weight, the variance less. Another task is Spoken Word Classification (speech recognition). The experimenters asked speakers to speak in five environments then trained the model

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with data. The performs of CW-RNN very good. Compared with standard RNN, LSTM, CW-RNN of parameters more detail, it can predict sound more accurately (RNN: 10,18,34,54,84, LSTM:5,8,17,26,41, CW-RNN:10,19,40,65,102). What's more, in the Error%, CW-RNN is the smallest. Experiments show that the CW-RNN can simplify the Standard RNN and improve the accuracy rate.

Two methods for CW-RNN are conceived, one is grouping hidden neurons into modules is a partway to having each weight have its clock. Still, it is not good, the other is running subsets of neurons at different speeds allows an RNN to efficiently learn the different dynamic time-scales inherent in complex signals, this method has been proved to be practicable and effective in the train.

B. W-RNN

Wang *et al*. proposed a concept of W-RNN (weighted RNN), which intends to mine the text [28]. The W-RNN can extract text serialization semantic information, which has a high commerce potential. Wang *et al* proved its suitability for text mining in the aspect of precision, recall, F1, and loss values. Text classification has three problems that ignore the relationship between semantic sparseness, and most algorithms cannot command time series change.

The W-RNN comprises three parts, long short-term memory (LSTM), WD-vector document representation, and NN classification. The author then implement W-RNN start from the text selection and use the formula (1)

$$SL = \min(DLS) \quad s.t. \theta \leq \frac{DNum(DL<SL)}{DNum} \tag{1}$$

This is used to calculate the fix length of the text. The author multiplies LSTM horizontality to form RNN to extract the text serialization information and then to classify the text. Vectors are inputted to LSTM, and word2vec produces a hidden state through these LSTM and cycle training in the above manner and utilizes a neural network to classify D-vector.

The formula of the calculation process of forget gate to calculate how much information will be discarded is (2)

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{2}$$

Input gate will decide which information needs to be update is shown in (3)

$$g_t = \sigma(W_g[h_{t-1}, x_t] + b_g) \tag{3}$$

and the information already update (4)

$$c_t = \sigma(W_c[h_{t-1}, x_t] + b_c) \tag{4}$$

The output gate is defined as Equation (5):

$$o_t = \sigma(W_t[h_{t-1}, x_t] + b_o) \tag{5}$$

h_t can capture information previous and current vocabulary, the calculation process is (6):

$$h_t = o_t * \tanh(s_t) \tag{6}$$

W-RNN changes the document vector based on RNN. Setting SL as the length of the document. Then train WM-vector, weight each WM-vector, and sum these vectors. Through these processes, the text category is known.

The author did an experiment to evaluate their method. The W-RNN is compared with four baselines to explain how their architecture W-RNN outperforms existing W-RNN. It compares with DNN, RNN, Bi-RNN, and CRNN. The experiment result is showing in this Table 1.

TABLE I. COMPARISON OF EXPERIMENTAL RESULTS BASED ON RNN TEXT CLASSIFICATION THE ADVANTAGES, DISADVANTAGES, AND IMPROVEMENTS OF ALL THE FOUR SELECTED METHODS.

	DNN	RNN	Bi-RNN	CRNN	W-RNN
Precision	56%	79%	75%	78%	85%
Recall	55%	77%	74%	76%	84%
F1	53%	78%	75%	77%	84%
Loss value	1.2	1.36	1.5	1.33	0.86

The W-RNN has better performance in all categories, which shows the ability to extract the deep meaning of the text. The accurate rate of classification of Bi-RNN is lower than RNN. CRNN extracts local mean and compresses the text data. RNN cannot extract semantic information from the text.

IV. DISCUSSION

This subsection discusses four representative works on RNN: CNN-RNN, ResNet-110, ResNet-164, and ResNet-1001, including their advantages, disadvantages, and how to improve the original RNN. The details are as follows:

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TABLE II. THE ADVANTAGES, DISADVANTAGES, AND IMPROVEMENTS OF ALL THE FOUR SELECTED METHODS.

	Advantages	disadvantages	improvements
CNN-RNN [29]	predicting the labels in a coarse-to-fine pattern. exploit the relationship between the coarse and fine categories.	It not available for many small datasets, so they need to examine taking advantage of traditional clustering methods towards automatically constructing a hierarchy for the objects.	the CNN-RNN generator utilizes the ground truth coarser-level labels as input and jointly optimizes the coarse and fine predictions, it means that Loss transform: use LSTM instead of standard RNN
Identity Mappings in Deep Residual Networks [30]	1, A new residual unit 2, 1000-layer deep networks	None.	1, When h (x) and f (y) take identity map, signal could be directly propagated from one unit to any other units, in both forward and backward passes. This makes training easier. 2, replacing the traditional post activate with pre activate
Deep Residual Learning for Image Recognition [31]	The residual learning framework alleviate the deep network training that hardly appear previously.	The result of testing 1202-layer network performs worse than 110-layer network because of overfitting.	Instead of learning unreferenced functions, reprogram the layer as a reference layer to input learning residual functions
Deep Networks with Stochastic Depth [32]	solve the problem of CNN of vanishing gradients, diminishing forward flow, slowly training time	Not mention in the article	It starts with a deep network. During the training process, for each small batch, a subset of a layer is randomly discarded and bypassed with an identity function

Table 3 shows the test error of different methods on the CIFAR-100 dataset.

TABLE III. THE TEST ERROR OF DIFFERENT METHODS ON THE CIFAR-100 DATASET.

Methods	Error
[29]	20.86%
[30]	24.58%
[31]	27.22%
[32]	22.71%

Table 4 shows the test error of different levels on the CIFAR-100 dataset for Resnet.

TABLE IV. THE TEST ERROR OF DIFFERENT LEVELS ON THE CIFAR-100 DATASET FOR RESNET

#layers	#params	Error
20	0.27M	8.75
32	0.46M	7.51
44	0.66M	7.17
56	0.85M	6.97
110	1.7M	6.43(6.61+-0.16)
1202	19.4M	7.93

V. FUTURE WORK

With the understanding of RNN get more and more mature, it will extend a lot of RNN-related algorithms. At that time, RNN will work as a basement for new algorithms. These algorithms will play a significant role in artificial intelligence. The future of RNN will like the root of a tree. Maybe future algorithms are no longer directly extended from RNN, but they have elemental RNN composition in their structure.

Although the RNN algorithm has been put forward since the 1980s, it is relatively old compared with other machine learning algorithms. Still, it has constantly been developing, making it has a strong vitality in both in-text and Machine translation. We think that RNNs can be combined with the standard algorithm in all fields of machine learning and can improve the effectiveness of the original algorithm, such as learn better features from unlabeled activity categories because of the characteristics of RNNs. This will make people carry out continuous research on it and will make further progress in the future.

VI. CONCLUSION

In this paper, we introduce the background of RNN and its theory, including its formula in several phases. By analyzing over 20 categories of RNN related algorithms, we give readers an idea of the main current application of RNN and in what direction RNN develops. In addition, combining with other algorithms enables RNN to show more possibilities in diverse fields. Then, through the qualitative and quantitative analysis in datasets CIFAR-100 and CIFAR-10, we propose the advantages and disadvantages of specific RNN-related algorithms. Moreover, we predict the future direction of RNN. We hope people can understand RNN, its usage, and its future development by doing these things.

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